

home: 7.8 km

[HOME]

Shah-rahe Sher Shah Suri Road
NORTH NAZIMABAD

five star

Allama Rasheed Torbani Road

landi kotal

aysha manzil

Shah-rahe Pakistan

7th St.

mukka chowk

Masoomen Road

Shah Jahany Road

water pump

Rizwan park

Ali Nuruzga Road

sorab goth

zia colony

Allama Shabbir Ahmael Usmani Road

Roshid

luckyone chorangi

gulshan chorangi

continental bakers

University Road

[UNIVERSITY]

nipa

GULSHAN-E-IPBAL

- five star: 7.5 km
- landi kotal: 5.9 km
- aysha manzil: 5 km
- mukka chowk: 4.8 km
- zia colony: 4 km

nagin chorangi

sakhi hassan

people chorangi

• nip: 4 km

• gulshan ch: 3.2 km

• luckyone ch: 5.5 km

• sorab goth: 8 km

• rizwan park: 5 km

• water pump: 5.2 km

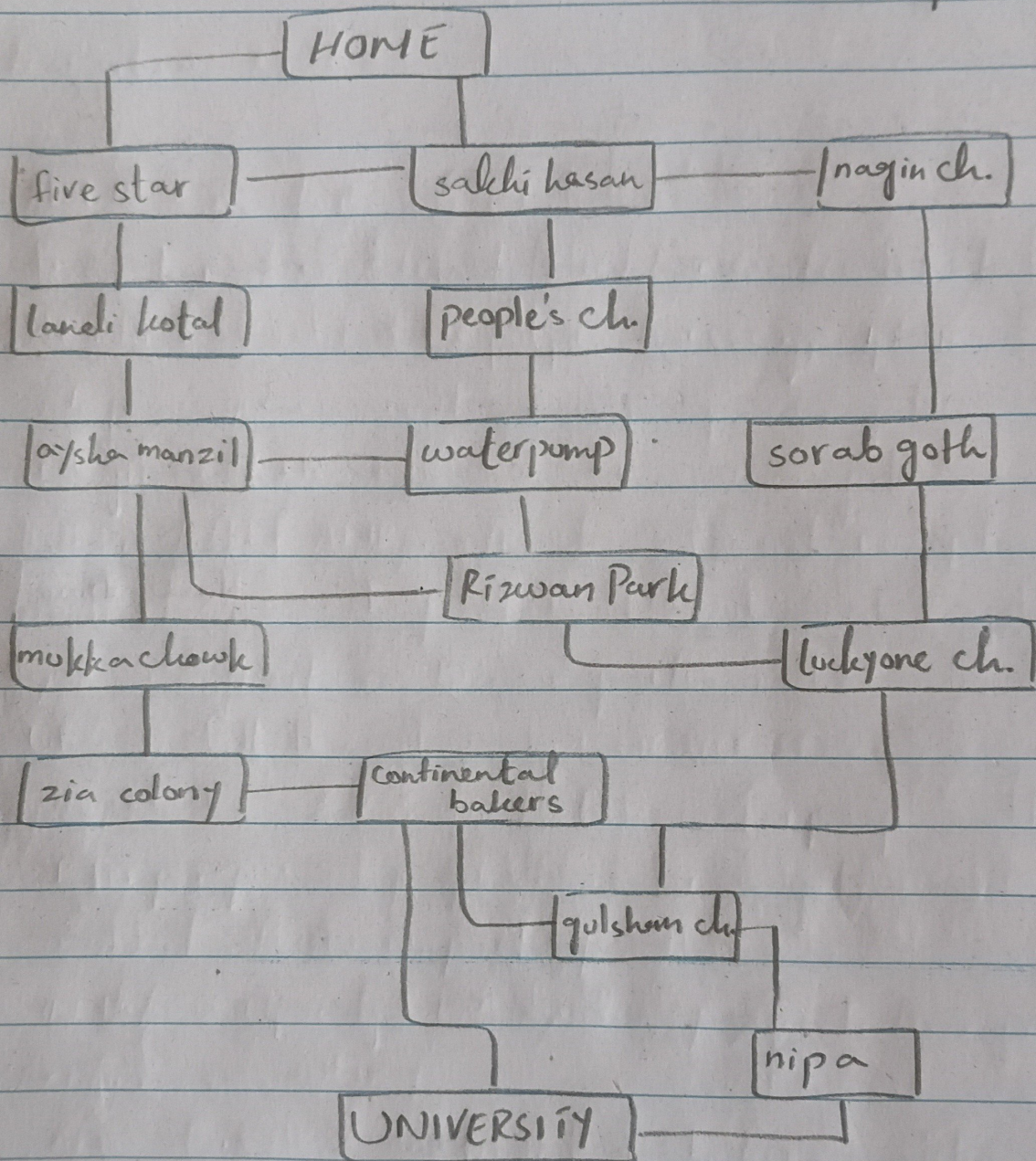
• people ch: 6 km

• sakhi hasan: 9 km

• nagin ch: 10 km

• continental: 3 km

Graph



Transition Table:-

- HOME → five star, sakhi hasan
- five star → sakhi hasan, home, laneli kotal
- sakhi hasan → five star, home, people's ch.,
nagin ch.
- nagin ch. → sakhi hasan, sorab goth
- laneli kotal → people ch., five star, ayshe
manzil.
- ayshe manzil → Rizwan park, mukka chowk,
laneli kotal, waterpump
- people ch → laneli kotal, waterpump, sakhi hasan
- waterpump → people ch, sorab goth, ayshe
manzil, Rizwan park
- mukka chowk → ayshe manzil, Rizwan
Park, zia colony
- zia colony → continental bakers, mukka
chowk.
- Continental bakers → gulshan ch., zia colony,
University
- sorab goth → nagin ch., waterpump, lucky one
ch.,
- Rizwan park → waterpump, ayshe manzil,

moleken chowk, luckyone ch.

- luckyone ch → golshan ch., sorab goth,
Rizwan park

- golshan ch → continental bakers, nipa,
luckyone

- nipa → University, golshan ch.

- University → nipa, continental bakers

(1)

A* Algo:-

Problem Home to Uni.

Home $\rightarrow$ None

PQ: [Home] V: [Home]

V: [Home]

Initialization: $f(n) = h(n) + g(n)$

State	$g(n)$	$h(n)$	$f(n)$
- Home	0	7.8	7.8

• Visit Home, Neighbours: [five star, sakhi hasan]

PQ: [five star, sakhi hasan]

V: [Home]

five star	1	7.5	8.5
sakhi hasan	1	9	10

Parent Table

Home $\rightarrow$ None, five star $\rightarrow$ Home

(2)

• Visit five star:

Neighbours: salehi hasan, laneli kotal, Home

- laneli kotal 2 5.9 7.9

PQ: [salehi hasan, laneli kotal]

V: [Home, five star]

Parent Table:

Home → None, five star → Home

laneli kotal → five star

• Visit laneli kotal

Neighbours: people's choranghi, aysa manzil, five star

PQ: [salehi hasan, aysa manzil, people ch.]

V: [Home, five star, laneli kotal]

- people's ch. 3 6 9

- aysa manzil 3 5 8

(3)

Parent Table:

Home → None, five star → Home

laneli kotal → five star

aysa manzil → laneli kotal

• Visit Aysa manzil

Neighbours: laneli kotal, waterpump, Rizwan Parli, mukka chowk

PQ: [salehi hasan, people ch, waterpump, Rizwan Parli, mukka chowk]

V: [Home, five star, laneli kotal, aysa manzil]

Rizwan Parli 4 5.1 9.1

waterpump 4 5.2 9.2

- mukka chowk 4 5 9

Parent Table

Home → None

five star → Home

laneli kotal → five star

aysa manzil → laneli kotal

mukka chowk → aysa manzil

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• Visit mukka chowk

Neighbours: aysha manzil, zia colony, Rizwan Park

PQ: [Salehi Hasan, people ch, water pump, Rizwan, zia]

V: [Home, five star, Laneli kotal, aysha manzil, mukka chowk]

State	glb	hlf	flf
zia colony	5	4	9

Parent Table

Home → None, five star → Home

Laneli kotal → five star

aysha manzil → Laneli kotal

mukka chowk → aysha manzil

zia colony → mukka chowk

• Visit Zia colony

Neighbours: mukka chowk, continental bakers

PQ: [Salehi Hasan, people ch, water pump, Rizwan, zia, continental bakers]

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V: [Home, five star, Laneli kotal, aysha manzil, Mukka chowk, zia colony]

State	glb	hlf	flf
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continental bakers	6	3	9
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Parent Table

Home → None, five star → Home

Laneli kotal → five star

aysha manzil → Laneli kotal

mukka chowk → aysha manzil

zia colony → mukka chowk

continental → zia colony

• Visit Continental:

Neighbours: University, golshan ch, zia colony

University	1	0	7
golshan ch	7	3.2	10.2

PQ: [sahli hasan, people ch. > water pump,
Rizwan Park, University, golshan ch.]
V: [Home, five star, laneli kotal, ayshe
manzil, mukka chowk, zia colony,
continental]

Parent Table

Home → None, five star → Home

laneli kotal → five star

ayshe manzil → laneli kotal

mukka chowk → ayshe manzil

zia colony → mukka chowk

continental → zia colony

University → continental

University → continental → zia colony →
mukka chowk → ayshe manzil → laneli
kotal → Home.

Home → laneli kotal → ayshe manzil →

mukka chowk → zia colony → continental
→ University.